

Group 7 Project - Milestone 4 (Due Friday, July 25)

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1. API Specification

/api/v1/search controller

Route 1: GET /api/v1/search/autocomplete

Description: Taking a search term as input, this endpoint returns an array of location matches. Location matches can be of 3 types: state, city, or US zip code. The search results array can include any of these 3 match types. The purpose of this endpoint is to populate autocomplete search results in our website's search bar.

Route Parameter(s): None

Query Parameter(s):

- term (string, required): the search term against which matches will be searched
- limit (int, default 3): the maximum number of matches that we want to return

Route Handler: SearchController.getAutocompleteSuggestions(term, limit)

Return Type: JSON object (with array included)

Return Parameters:

```
{
  "success": (boolean),
  "data": {
    "suggestions": [
      {
        "value": (string) ex. "12301",
        "matchtype": (string) ex. "ZipCode"
      },
      {
        "value": (string) ex. "New York, NY",
        "matchtype": (string) ex. "CityAndState"
      },
      {
        "value": (string) ex. "NY",
        "matchtype": (string) ex. "State"
      }
    ]
  },
  "timestamp": (string) ex. "2025-07-22T13:10:46.834Z",
  "type": "autocomplete_suggestions"
}
```

COMPLEX ROUTE (~5s)

Route 2: GET /api/v1/search/featured-cities

Description: This endpoint returns an array of cities that have some of the best metrics around their housing prices and health measures. Its purpose is to highlight a shortlist of cities on our homepage.

Route Parameter(s): None

Query Parameter(s):

- limit (int, default 3): the maximum number of featured cities that we want to return

Route Handler: SearchController.getFeaturedCities(limit)

Return Type: JSON object (with array included)

Return Parameters:

```
{
  "success": (boolean),
  "data": {
    "cities": [
      {
        "city": (string) ex. "Bassett",
        "state": (string) ex. "NE",
        "population": (string) ex. "109695",
        "avg_listing_price": (string) ex. "95948.09",
        "avg_health_measure": (string) ex. "0.30",
        "price_rank": (string) ex. "1811",
        "health_rank": (string) ex. "2246",
        "combined_rank": (string) ex. "4057"
      }
    ]
  },
  "timestamp": (string) ex. "2025-07-22T13:20:11.918Z",
  "type": "featured_cities"
}
```

COMPLEX ROUTE (~2min)

Route 4: GET /api/v1/search/zipcode-summaries

Description: This endpoint returns an array of ZIP codes with summary data about each of them—with multiple query parameters that filter that list. Each row will be used to display a (1) result card using that summary data and (2) dot on the interactive map using latitude and longitude.

Route Parameter(s): None

Query Parameter(s):

- state (string, required): state to filter for ZIP codes
- city (string, optional): city to filter for ZIP codes
- limit (int, default 10): the maximum number of results to return
- minPopulation (int, optional)
- maxPopulation (int, optional)
- minHousingPrice (int, optional)
- maxHousingPrice (int, optional)
- minAverageIncome (int, optional)
- maxAverageIncome (int, optional)
- maxHealthMeasure (int, optional)
- healthMeasure (string, optional): health measure to check against maxHealthMeasure
- minPoliceDepartments (int, optional)
- minPoliceOfficers (int, optional)

- minHospitals (int, optional)
- minFirefighterStations (int, optional)
- minFirefighters (int, optional)
- minChildcareCenters (int, optional)

Route Handler: SearchController.getZipcodeSummaries(<all filters here>)

Return Type: JSON object (with array included)

Return Parameters:

```
{
  "success": (boolean),
  "data": [
    {
      "zip_code": (string) ex. "12345",
      "city": (string) ex. "New York",
      "State": (string) ex. "NY",
      "population": (string) ex. "109695",
      "median_price": (string) ex. "95948.09",
      "mean_income": (string) ex. "95948.09",
      "health_ratio": (string) ex. "0.30",
      "police_departments_count": (string) ex. "1811",
      "Num_police_officers_count": (string) ex. "2246",
      "hospitals_count": (string) ex. "4057",
      "fire_stations_count": (string) ex. "1234",
      "firefighters_count": (string) ex. "543",
      "childcare_centers_count": (string) ex. "987"
    }
  ],
  "timestamp": (string) ex. "2025-07-22T13:20:11.918Z",
  "type": "zipcode_summaries"
}
```

/api/v1/health controller

Route 5: GET /api/v1/health/measures

Description: This endpoint returns an array of available health measures that our database stores (as well as some summary statistics about those health measures). Its primary purpose is to populate a select dropdown in our search results page, so users can filter by the health measures they care about.

Route Parameter(s): None

Query Parameter(s): None

Route Handler: HealthController.getAvailableHealthMeasures()

Return Type: JSON object (with array included)

Return Parameters:

```
{
  "success": (boolean),
  "data": [
    {
      "measure": (string) ex. "Current asthma among adults",
      "measure_count": (string) ex. "32520",
      "min_population": (int) ex. 52,

```

```

    "max_population": (int) ex. 135256
  }
],
"timestamp": (string) ex. "2025-07-22T13:27:11.674Z",
"type": "available_health_measures"
}

```

Route 8: GET api/v1/health/community/:zipcode

Description: Returns community health properties as fractional number between 0 and 1 calculated as percentage (asthma, obesity, social isolation etc.) along with the total number of hospitals and hospital beds per population for a given zipcode ZIP. (Note: we can reuse it for the Comparison Page)

Route parameters:

- zipcode (string) (path, required): 5-digit zip code to get the data.

Query Parameter(s): None

Route Handler: (req, res) => HealthController.getCommunityHealthProperties(zipcode);

Return Type: JSON Object (with array included)

Return Parameters:

```

{
  "success": (boolean) ex. true,
  "data" (array): [
    {
      "asthmarate": (float) ex. 0.37
      "obesityrate": (float) ex. 0.81
      "doctorvisitsrate": (float) ex. 0.912
      "depressionrate":(float) ex. 0.213
      "housinginsecurityrate": (float) ex. null
      "socialisolationrate": (float) ex. null
      "hospitalcount": (string) ex. "1"
      "hospitaltopopulationratio": (string) ex. "8119.00"
      "bedstopopulationratio": (string) ex. "31.00"
    }
  ],
  "timestamp": (string) ex. "2025-07-24T01:06:51.887Z",
  "type": (string) ex. "community_health_properties"
}

```

/api/v1/real-estate controller

Route 7: GET api/v1/real-estate/price-trends/:zipcode

Description: Returns the median listing price and corresponding month date for a given zipcode.

Route parameters:

- zipcode (string) (path, required): 5-digit zip code to get the data.

Query Parameter(s): None

Route Handler: RealEstateController.getPriceTrends(zipcode)

Return Type: JSON Object (with array included)

Return Parameters:

```
{
  "success": (boolean),
  "data" (array): [
    {
      "monthdate": (string) ex. "77001",
      "medianlistingprice": (int) ex. 21000
    },
    "timestamp": (string) ex. "2025-07-24T01:02:46.110Z",
    "type": (string) ex. "price_trends",
    "zipcode": (string) ex. "19104"
  ]
}
```

Route 13: GET /api/v1/real-estate/lowest-price-zipcode/:city/:state

Description: This API uses city(string) and state(string) as input. This API will find the lowest home price zipcode within the input city and state. It will filter out outlier data with abnormally low home prices. This API will be used in the comparison page to populate relevant information for the input city.

Route Parameter(s):

- City (string) (path, required): name of the city,
- State (string) (path, required): abbreviation of the State

Query Parameter(s): None

Route Handler: realEstateController.getZipcodeWithLowestMedianHomePriceInCity(city, state)

Return Type: JSON object (with array included)

Return Parameters:

```
{
  "success": (boolean),
  "data": {
    "zipcode": (string) ex. "77001",
    "median_home_price": (int) ex. 49900,
    "city": (string) "Houston",
    "state": (string) "TX",
    "rank": (string) "1"
  },
  "timestamp": (string) "2025-07-23T23:43:21.754Z",
  "type": (string) "lowest_median_home_price_zipcode",
  "city": (string) "Houston"
}
```

/api/v1/analytics controller

Complex Route ~15 seconds

Route 12: GET /api/v1/analytics/similar-cities/:city/:state

Description: The input to this api is city(string) and state(string). This API will find and return the top 2 U.S. cities most similar to a given city based on multiple socio-economic and health factors. The similarity is computed using a weighted scoring function across metrics such as: Population, Mean income, Median listing price, Number of police/fire departments and personnel, Number of hospitals and childcare centers. This API will be used in the comparison page to suggest similar cities to review.
, health indicators (asthma, obesity, depression)

Route Parameter(s):

- City (string) (path, required): name of the city
- State (string) (path, required): abbreviation of the State

Query Parameter(s): None

Route Handler: AnalyticController.getCityAggregateComparison(city, state)

Return Type: JSON object (with array included)

Return Parameters:

```
{
  "success": (boolean)
  "data":{
    "sourceCity": (string) ex. "Houston",
    "sourceState": (string) ex. "TX",
    "similarCities":[
      {
        "city": (string) ex. "Chicago",
        "state": (string) ex. "IL",
        "similarityScore": (int) ex. 26255
      },
      {
        "city": (string) ex. "San Antonio",
        "state": (string) ex. "TX",
        "similarityScore": (int) ex. 72859
      }
    ],
    "timestamp": (string) ex. "2025-07-23T22:14:33.620Z",
    "type": (string) ex. "similar_cities"
  }
}
```

Route 10: GET /api/v1/analytics/city-aggregate-comparison

Description: This API takes city1 and state1 as the input. This api will find the summarized result of police, health, childcare and fire department results for each city and state. The purpose of this endpoint is to collect basic information for each city in the comparison page.

Route Parameter(s): None

Query Parameter(s):

- city1 (string) (required): name of the city
- state1 (string) (required): abbreviation of the State

Route Handler: AnalyticController.getCityAggregateComparison(city, state)

Return Type: JSON object (with array included)

Return Parameters:

```
{
  "success": (boolean),
  "data":{
    "city": (string) ex. "Houston",
```

```

    "state": (string) ex. "TX",
    "totalPopulation": (int) ex. 3198284,
    "meanIncome": (float) ex. 103131.95833333333,
    "numFireDept": (int) ex. 18,
    "numFirefighters": (int) ex. 5149,
    "numPoliceDept": (int) ex. 58,
    "numPoliceOfficers": (int) ex. 8664,
    "numChildCareCenter": (int) ex. 943,
    "numHospitals": (int) ex. 68,
    "obesityRate": (float) ex. 0.35653608247422686
  },
  "timestamp": (string) ex. "2025-07-23T22:50:25.578Z",
  "type": (string) ex. "city_aggregate_comparison"
}

```

Route 11: GET /api/v1/analytics/city-growth-rate/:city/:state

Description: The inputs of this API are City (string) and state (String). This endpoint returns the growth rate of the desired city. We will be able to run this query twice for the comparison between two city, state pairs. This API will be used in the comparison page to help compare the two cities.

Route Parameter(s):

- City (string) (path, required): name of the city
- State (string) (path, required): abbreviation of the State

Query Parameter(s): None

Route Handler: AnalyticController.getCityGrowthRate(city, state)

Return Type: JSON object (with array included)

Return Parameters:

```

{
  "success": (boolean),
  "data":
  {
    "city": (string) ex. "Houston",
    "state": (string) ex. "TX",
    "growthRate3Year": (string) ex. "-4.84%",
  },
  "timestamp": (string) ex. "2025-07-23T21:55:08.071Z",
  "type": (string) ex. "city_growth_rate"
}

```

Route 6: GET /api/v1/analytics/location/:zipcode

Description: Returns the full description such as ZipCode, City, State, Population, MeanIncome, MedianListingPrice, ActiveListingCount, PoliceStations, PoliceOfficers, number of Hospitals, number of ChildcareCenters, number of FirefighterDepartments, number of Firefighters for a given zip code. This data is required to fill the data in the location page for a given zipcode ZIP.

Route parameters:

- zip code (string) (path, required): 5-digit zip code to get the data

Query Parameter(s): None

Route Handler: analyticsController.getLocationPageData(zipcode)

Return Type: JSON Object (with array included)

Return Parameters: {

```
"success": (boolean) ex.true,
  "data" (object) : {
    "zipcode": (string) ex. "77001",
    "location": {
      "city": (string) ex. "Philadelphia",
      "state": (string) ex. "PA"
    },
    "demographics": {
      "population": (int) ex. 7100
    },
    "realEstate": {
      "medianPrice": (int) ex. 12000
      "activeListings": (int) ex. 14
    },
    "income": {
      "meanIncome": (int) ex. 1400
    },
    "facilities": {
      "police": {
        "stations": (int) ex. 12
        "officers": (int) ex. 12
      },
      "hospitals": (int) ex. 1
      "fire": {
        "departments": (int) ex. 1
        "personnel": (int) ex. 12
      },
      "childcare": (int) ex. 1
    },
    "timestamp": (string), ex. "2025-07-23T21:55:08.071Z"
  },
  "timestamp": (string) ex. "2025-07-23T21:55:08.071Z",
  "type": (string) ex. "location_page_data",
}
```

Complex Route ~8 seconds

Route 9: GET api/v1/analytics/similar-zipcodes/:zipcode

Description: Returns 4 most similar **zipcodes** for the given ZIP code based on the similarity score which is measured as the sum of absolute difference between different attributes such as population, meanincome, average listing price, services (total number of firefighters, number of fire departments, total number of police officers, number of police stations number

of childcare centers), and health measures (Current Asthma among adults, Obesity among Adults, Housing Insecurity in the past 12 months among adults, Feeling socially isolated among adults). Each attribute is assigned a weighted value between 0 and 1 in calculations.

Route parameters:

- zipcode (string) (path, required): 5-digit zip code to get the data.

Query Parameter(s): None

Route Handler: AnalyticsController.getSimilarZipcodes(zipcode);

Return Type: JSON Object (with array included)

Return Parameters:

```
{
  "success": (boolean),
  "data" (object): {
    "sourceZipcode": (string) ex. "19104"
    "similarZipcodes": (array) [
      {
        "zipcode": (string) ex. "89030"
        "similarityScore": (int) ex. 347
      },
      {
        "zipcode": (string) ex. "89031"
        "similarityScore": (int) ex. 476
      },
      {
        "zipcode": (string) ex. "71121"
        "similarityScore": (int) ex. 1203
      },
      {
        "zipcode": (string) ex. "71123"
        "similarityScore": (int) ex. 1702
      }
    ],
    "analysis": {
      "method": (string) ex. "weighted attributes",
      "count": (int) ex. 4
    }
  },
  "timestamp": (string) ex. "2025-07-24T01:04:20.300Z",
  "type": (string) ex. "similar_zipcodes"
}
```

Route 3: GET /api/v1/analytics/homepage-featured-zipcodes

Description: This API will return an array of zip code along with relevant information on the number of facilities. Each zip code will have useful information such as facilities, income, and

score. The purpose of this endpoint is to populate our home page with featured zip codes that have good access to social service.

Route Parameter(s): None

Query Parameter(s):

- limit (int ,optional): how many zipcodes to return for result. The default value is 4.

Route Handler: AnalyticController.getHomepageFeaturedZipcodes(limit)

Return Type: JSON object (with array included)

Return Parameters:

```
{
  "success": (boolean)
  "data" (object) :[{
    "zipcode": (string) ex. "10013",
    "facilities":{
      "hospitals": (int) ex. 0,
      "police": (int) ex. 11,
      "firefighter": (int) ex. 0,
      "total": (int) ex. 11
    },
    "healthMetrics":{
      "avgPoorHealthRatio": (float) ex. 17,
      "healthMeasuresCount":4,
      "population":31042
    },
    "income":{
      "meanIncome":385678
    },
    "scores":{
      "staffingServiceScore": (float) ex. 1,
      "healthServiceScore": (float) ex. 0.825,
      "serviceLevelScore": (float) ex. 0.183,
      "rank": (int) ex. 15759
    },
    "analysis":{
      "serviceLevel": (string) ex. "Critical Service Level",
      "healthStatus": (string) ex. "Good Health Outcomes",
      "facilityStatus": (string) ex. "Well Served"
    }
  }
}],
  "timestamp" (string) ex. : "2025-07-25T13:59:18.070Z",
  "type": (string) ex. "homepage_featured_zipcodes",
  "limit": (int) ex. 4
}
```